ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 4000 J	400 W x 10 s.
3	**(b)** algal bloom and fish kill	Eutrophication.
4	**(a)** the ground is heated by sunlight and warms the air	Surface heating.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(b)** increase	More absorption.
7	**(b)** water filtration and flood control	Natural sponge.
8	**(b)** absorption of UV by ozone	Ozone heating.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** 100	2200/22.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorbed = 0.7 x 342 (1); = 239 W/m squared (1).	1+1
12	Long coastline and monsoon dependence (1); Large population in agriculture (1).	1+1
13	UV has the energy to break O2 and O3 bonds (1); Visible light has less energy (1).	1+1
14	Ice-albedo feedback (1); Release of methane from thawing permafrost (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Annual emission = 250 x 12 x 0.8 = 2400 kg (1); Trees = 2400/22 (1); = 109 trees (1).	1+1+1
16	Dry air has little water vapour to trap heat (1); Quick cooling at night (1); Bare sand heats up fast (1).	1+1+1
17	30 units are reflected and 70 are absorbed (1); The Earth radiates back 70 units in equilibrium (1); Greenhouse gases delay the escape of heat (1).	1+1+1
	OR: Saving = 20 W x 100 = 2 kW (1); Energy = 2 x 8 x 250 = 4000 kWh (1); CO2 = 3200 kg (1).	1+1+1
18	Absorb CO2 and release O2 (1); Respiration uses much of the oxygen too (1); Better described as carbon stores and biodiversity hotspots (1).	1+1+1
	OR: Releases carbon stored for millions of years (1); Natural sinks cannot absorb all (1); CO2 accumulates (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Absorbed = 0.7 x 342 = 239 W/m squared (1); At equilibrium the outgoing is 239 W/m squared (1); Albedo 0.28: absorbed 246 W/m squared, warming (1); More greenhouse gases reduce the outgoing radiation (1); Surface warms until balance is restored (1).	1+1+1+1+1
	OR: Less transpiration, less rainfall (1); Carbon released (1); Albedo changes (1); Biodiversity loss (1); Soil erosion and floods (1).	1+1+1+1+1
20	Limestone heated: CaCO3 gives CaO + CO2 (1); Fuel burning adds CO2 (1); Cement 0.9 t CO2 per tonne (1); Recycling steel uses less energy (1); Renewable power and efficiency lower emissions (1).	1+1+1+1+1
	OR: Volume = 0.8 x 20 x 10^6 = 1.6 x 10^7 cubic m (1); Harvestable = 9.6 x 10^6 cubic m (1); More than the demand of 5 x 10^6 (1); Storage and recharge required (1); Reduces dependence on groundwater (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** its increase beyond natural levels enhances the greenhouse effect	Balance matters.
2	**(b)** weather and communication	Space technology.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(d)** 1.8 tonnes	12000 x 150 g.
5	**(c)** reducing the use of chemical fertiliser	Fertiliser production emits CO2.
6	**(b)** 750 s	d/c.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** 15 kWh	0.1 x 150.
9	**(c)** air pollution and loss of soil nutrients	Smoke and CO2.
10	**(d)** 960 W/m squared	70 per cent.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The 100-year gas (1); Cuts in long-lived gases are urgent (1).	1+1
12	Water absorbs most radiation (1); Dark appearance (1).	1+1
13	Seasonal photosynthesis changes uptake (1); Rising trend is due to fossil fuels (1).	1+1
14	Net emission equals absorption (1); A community planting trees to offset emissions (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	CO2 dissolves forming carbonic acid (1); pH of the sea falls (1); Harms corals and shellfish (1).	1+1+1
16	420 + 50 = 470 ppm (1); Increase 50/420 (1); = 11.9 per cent (1).	1+1+1
17	Reservoir that absorbs more carbon than it releases (1); Forests, oceans, soil (1); Carbon capture and storage (1).	1+1+1
	OR: Excess nitrate runs off (1); Eutrophication (1); Organic manure and crop rotation with legumes reduce use (1).	1+1+1
18	Land heats faster than the sea in summer (1); Moist sea air flows in and gives rain (1); Warmer oceans give more moisture and heavier but erratic rainfall (1).	1+1+1
	OR: Area needed to support a person's consumption (1); Compare by diet, travel, energy use (1); Reduction by LiFE habits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cool roofs with high albedo (1); Trees and parks give shade and transpiration (1); Water bodies cool by evaporation (1); Efficient buildings reduce heat from air conditioners (1); Lower temperatures reduce heat stress and electricity demand (1).	1+1+1+1+1
	OR: Ozone: few chemicals, alternatives available (1); Climate: CO2 from all economy (1); Both global problems needing agreements (1); Montreal succeeded with trade controls (1); Climate needs deeper changes (1).	1+1+1+1+1
20	Two identical containers: black and white (1); Same light, same time (1); Measure temperature rise (1); Black rises more (1); Link: albedo of snow, forest, deserts (1).	1+1+1+1+1
	OR: Ice cores trap ancient air (1); Glacial: 180 ppm, cold (1); Pre-industrial: 280 ppm (1); Today: 420 ppm, faster change (1); Human activity as the cause (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 12 kg	15 kWh x 0.8.
3	**(a)** melting ice lowers albedo and increases warming	Amplifies the change.
4	**(b)** 4	A sphere has 4 times the area of its cross-section.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(b)** 90 per cent	90/100.
7	**(b)** more CO2 helps plants grow and absorb more CO2	Dampens the change.
8	**(a)** 342 W/m squared	1370/4.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(d)** 550 tonnes	150 x 3.67.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Advantages: absorbs CO2, restores habitat (1); Limitations: takes years, land needed (1).	1+1
12	44/12 x 1 (1); = 3.67 kg (1).	1+1
13	Warm water expands (1); Melting land ice adds water (1).	1+1
14	UV damages phytoplankton (1); Affects marine food chains and CO2 absorption (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	C + O2 gives CO2: 12 g gives 44 g (1); CO2 = 6 x 44/12 = 22 kg (1); Oxygen used = 16 kg (1).	1+1+1
16	CO2 dissolves in rainwater forming acid (1); Reacts with rocks (1); Carbonates carried to the sea (1).	1+1+1
17	Photosynthesis in oceans gives about half of the oxygen (1); They absorb CO2 (1); Base of the marine food chain (1).	1+1+1
	OR: Atmosphere is mostly transparent to visible light (1); Surface absorbs and warms (1); Surface heats the air by conduction and infrared (1).	1+1+1
18	CO2: fossil fuels, deforestation (1); CH4: paddy fields, cattle, landfills (1); Methane traps more heat per molecule but is short-lived (1).	1+1+1
	OR: Changes in weather patterns (1); Loss of polar bear habitat (1); New shipping routes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Short-wave radiation passes (1); Long-wave radiation absorbed by gases (1); Re-emitted downward (1); Water vapour, CO2, methane (1); Surface about 33 degrees Celsius warmer (1).	1+1+1+1+1
	OR: Energy: LED, solar panels (1); Transport: carpooling, cycling (1); Waste: segregation, composting (1); Water: harvesting and recycling (1); Trees to offset and monitoring (1).	1+1+1+1+1
20	Ocean uptake = 3 Gt (1); Land uptake = 2.5 Gt (1); Atmosphere gains 4.5 Gt (1); Rise = 4.5/2.1 = 2.1 ppm (1); Shows the need to cut emissions (1).	1+1+1+1+1
	OR: Warm oceans expand and evaporate more (1); Atmosphere: stronger cyclones, changed rainfall (1); Geosphere: coastal erosion and flooding (1); Biosphere: coral bleaching, fish migration (1); Feedback to the atmosphere (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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